THE TRADING DESK THAT SEES CONTAGION BEFORE PRICES DO

Knowledge Graphs, Obsidian Vaults, Evidence Retrieval and Bounded AI Judgment

Alejandro Reynoso | October 2026

The core proposition

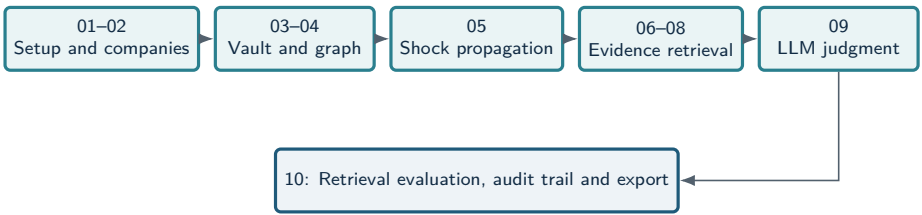
A Second Brain is not just a collection of notes.

It is a governed architecture that connects evidence, models and judgment.

VAULT	GRAPH	SIMULATION	JUDGMENT
Corporate dossiers	Typed links	Stress pathways	Reviewable memo

Research question Can graph-grounded evidence support a more reasoned institutional assessment of indirect vulnerability than unordered document retrieval?

The architecture: ten stages, one evidence chain



Knowledge is selected and interpreted only after its lineage and network relevance are established.

Cell 01 | Environment, reproducibility and secrets

PEDAGOGICAL OBJECTIVE

Establish the runtime, working directory, random seed and API configuration.

- **Reproducibility:** fixed synthetic universe and auditable outputs.
- **Storage:** persistent Google Drive working directory.
- **Model:** Claude Haiku 5.5 only when an API key is available.
- **Governance:** a failed API call is not a generated judgment.

Key lesson A credible Second Brain begins with reproducible computation, not with a prompt.

Cell 02 | The 500-company synthetic universe

PEDAGOGICAL OBJECTIVE

A controlled laboratory where sector, region and financial characteristics are known.

- 500 fictional corporate entities across 10 sectors and 5 regions.
- Each has revenue, leverage, cash-buffer and descriptive metadata.
- Balanced sector counts make comparisons teachable.
- The registry is the node-level source of record.

Key lesson Registry data make a company identifiable; they do not explain interdependence.

Cell 03 | Typed relationships and the Obsidian vault

PEDAGOGICAL OBJECTIVE

Turn separate corporate records into interconnected, evidence-linked institutional memory.

- Three directed edge types: supplies, finances, owns_stake_in.
- Each edge records weight, evidence ID, confidence and validity interval.
- Markdown dossiers describe incoming and outgoing relationships.
- Obsidian links such as `[[CO_001]]` support navigation.

Key lesson Knowledge becomes useful when connections and provenance are preserved.

Cell 04 | The computational knowledge graph

PEDAGOGICAL OBJECTIVE

Separate the full institutional relationship map from operational supply transmission.

- `MultiDiGraph`: preserves multiple typed edges between companies.
- `DiGraph SUP`: contains supplier-to-customer links only.
- Source and target IDs, evidence uniqueness and weights are validated.
- Local subgraphs reveal how dependencies can cross sectors.

Key lesson A link means something only when its economic type and direction are explicit.

Cell 05 | Modeling and propagating a shock

PEDAGOGICAL OBJECTIVE

Trace an East Asian semiconductor disruption through the directed supply network.

- Four seeded manufacturers; synthetic initial severity 0.90.
- At each hop: $x_v = x_u w_{uv}(0.62)$.
- Maximum three hops; no cycles within a path; discard values below 0.025.
- Retain the strongest exposure path, not summed systemic loss.
- Risk indicator adjusts for leverage and cash buffer.

Key lesson A network-stress score is neither a forecast nor a default probability.

Cell 06 | Baseline A and B: context selection

PEDAGOGICAL OBJECTIVE

Compare a bare event description with sequential, document-rich retrieval.

- A: shock event, affected IDs and analytical question only.
- B: append Obsidian dossiers in registry order.
- Stop when the 12,000-character limit is reached.
- More pages do not ensure inclusion of relevant pathways.

Key lesson An evidence-rich archive can still produce an evidence-poor prompt.

Cell 07 | Baseline C: TF–IDF retrieval

PEDAGOGICAL OBJECTIVE

Rank companies by lexical relevance to a semiconductor-disruption query.

- Represent each company through name, sector, region and description.
- TF–IDF unigrams and bigrams; cosine similarity.
- Select 25 company IDs and abbreviated Markdown excerpts.
- Lexical similarity measures topic match, not supply dependence.

Key lesson A company can be textually distant and economically exposed.

Cell 08 | Strategy D: graph-aware retrieval

PEDAGOGICAL OBJECTIVE

Pass the model actual modeled transmission routes, not merely relevant descriptions.

- Remove four directly shocked firms from downstream discovery.
- Rank downstream firms by the synthetic risk indicator.
- Select 25 companies and supply detailed paths for the first 15.
- Attach company IDs, edge evidence IDs and modeled exposure.
- Save the top three pathways per selected company for audit.

Key lesson The LLM receives a traceable explanation of why a company is exposed.

Cell 09 | Where knowledge meets bounded judgment

PEDAGOGICAL OBJECTIVE

Claude converts an evidence package into a reviewable institutional market-risk memorandum.

- System instruction: use supplied evidence only; never invent IDs or prices.
- Five memo sections: decision summary, pathways, controls, uncertainty, review.
- Suggest analytical safeguards, never executable trades.
- Record model status; no response is not a substitute judgment.

Key lesson The graph calculates. The model interprets. People remain accountable.

Cell 10 | Evaluation and reproducible export

PEDAGOGICAL OBJECTIVE

Measure the retrieval pipeline and preserve the complete laboratory.

- Benchmark: top 20 downstream firms under the synthetic risk model.
- Compare selected firms, recall@20, context size and source density.
- Graph-aware retrieval uses the same model as the benchmark: not independent proof.
- Save CSV, JSON, contexts, README and a reproducible ZIP archive.

Key lesson Retrieval recall measures evidence selection, not correctness of AI judgment.

Why graph-aware retrieval is analytically different

Evidence method	Selection rule	What it cannot establish
A: Event only	Question and affected firms only	Firm-level pathway evidence
B: Broad notes	First available dossiers under budget	Priority of structural links
C: TF–IDF	Lexical match to disruption terms	Actual downstream dependence
D: Graph-aware	Risk-ranked downstream entities and evidence paths	Empirical predictive validity

Interpretation The graph-aware package is better aligned to modeled propagation; its superiority is not independently validated by this synthetic exercise.

What the LLM adds—and what it must not do

ADDED CAPABILITIES

- Synthesize pathways into a coherent risk narrative.
- Explain sector and corporate implications.
- Suggest monitoring and portfolio safeguards.
- Make uncertainty and missing evidence explicit.

BOUNDARIES

- No invented company or evidence identifiers.
- No fabricated prices or historical statistics.
- No treating stress scores as probabilities.
- No autonomous execution of trades.

Human governance The model is an evidence-constrained analytical interface, not an autonomous portfolio manager.

What Laboratory 02 establishes

1. A typed, directed knowledge graph makes indirect corporate dependencies navigable.
2. Obsidian Markdown dossiers provide readable institutional memory with source links.
3. Deterministic stress propagation can produce auditable synthetic exposure pathways.
4. Evidence selection changes the quality and economy of the information given to the LLM.
5. Bounded model judgment can turn graph evidence into a structured memo for human review.

Methodological limit All companies, edges, shocks and metrics are synthetic. This is a demonstration of architecture and research method, not a real-world market forecast.

SECOND BRAIN | LABORATORY 02

FROM FRAGMENTED INFORMATION TO EVIDENCE-GROUNDED JUDGMENT

Knowledge graphs + Obsidian vaults + deterministic analysis
Bounded LLM judgment + institutional governance

The Second Brain Project | Part of a five-laboratory series